Field extensions

Chapter 1. Field extensions

§1.1 Fields

Example 1.1. The following are examples of fields.
- The field of rational numbers $\mathbb{Q}$.
- The field of real numbers $\mathbb{R}$.
- The field of complex numbers $\mathbb{C}$.
- For any prime number p, the ring $\mathbb{Z}/p\mathbb{Z}$ of integers modulo p is a field, denoted by $\mathbb{F}_p$.
- For any prime number p and positive integer n, there exists a unique (up to isomorphism) finite field of order p^n, denoted by $\mathbb{F}_{p^n}$.

Exercise 1.2. Show that the following are not fields.
- The ring of integers $\mathbb{Z}$.
- The ring of polynomials $\mathbb{Q}[x]$ in one variable x over $\mathbb{Q}$.
- The ring of polynomials $\mathbb{R}[x, y]$ in two variables x, y over $\mathbb{R}$.

Exercise 1.3. Show that any finite integral domain is a field.

Exercise 1.4. Show that any finite field has order p^n for some prime number p and positive integer n.

Example 1.5. For any field k, the ring of polynomials in one variable x over k is denoted by $k[x]$. The field of rational functions in one variable x over k is denoted by $k(x)$, and it consists of all elements of the form $\frac{f(x)}{g(x)}$, where $f(x), g(x) \in k[x]$ and $g(x) \neq 0$. The field $k(x)$ is the field of fractions of the integral domain $k[x]$, and it is called the **field of rational functions** in one variable x over k.

Example 1.6. For any field k, the **ring of formal power series** in one variable x over k is denoted by $k[[x]]$. The **field of formal Laurent series** in one variable x over k is denoted by $k((x))$, and it consists of the generalized formal power series, that is, all elements of the form

$$\sum_{n \geq n_0} a_n x^n$$

for some integer n_0, where $a_n \in k$ for all $n \geq n_0$ and $a_{n_0} \neq 0$. The field $k((x))$ is the field of fractions of the integral domain $k[[x]]$, and it is called the **field of formal Laurent series** in one variable x over k.

§1.2 Field homomorphisms and isomorphisms

Definition 1.7. Let K and L be fields. A **field homomorphism** from K to L is a map $\sigma : K \to L$ that preserves the field operations, that is, for all $a, b \in K$, we have

- $\sigma(a + b) = \sigma(a) + \sigma(b)$,
- $\sigma(ab) = \sigma(a)\sigma(b)$,
- $\sigma(1_K) = 1_L$, where 1_K and 1_L denote the multiplicative identities of K and L, respectively.

A homomorphism $\sigma : K \to L$ is called an **isomorphism** if there exists a homomorphism $\tau : L \to K$ such that $\tau \circ \sigma = \mathrm{id}_K$ and $\sigma \circ \tau = \mathrm{id}_L$, where id_K and id_L denote the identity maps on K and L, respectively.

Exercise 1.8. Show that if $\sigma : K \to L$ is a field homomorphism, then σ is injective. Is it true that σ is surjective? If yes, prove it. If not, give a counterexample.

Exercise 1.9. Let $\sigma : K \to L$ be a field homomorphism. Show that σ is an isomorphism if and only if σ is surjective.

§1.3 Extension of fields

Definition 1.10. Let K be a field. A **field extension** of K is a field L that contains K as a subfield. We often denote a field extension by L/K, which indicates that L is an extension of K. The field K is called the **base field**, and the field L is called the **extension field**.

Example 1.11.

- The field of complex numbers $\mathbb{C}$ is an extension of the field of real numbers $\mathbb{R}$.
- Similarly, the field of real numbers $\mathbb{R}$ is an extension of the field of rational numbers $\mathbb{Q}$.
- For any field k, the field of rational functions $k(x)$ in one variable x over k is an extension of k.

- For any field k, the field of formal Laurent series $k((x))$ in one variable x over k is an extension of k.

Definition 1.12. Let L/K be a field extension. An **intermediate extension** of L/K is a field M such that $K \subseteq M \subseteq L$. In this case, we say that M/K is a subextension of L/K.

Example 1.13. For any field k, the field of rational functions $k(x)$ in one variable x over k is an extension of k. The field of rational functions $k(x^2)$ in one variable x^2 over k is an intermediate extension of $k(x)/k$.

Example 1.14. For any field k, the field of formal Laurent series $k((x))$ in one variable x over k is an extension of k. The field of formal rational functions $k(x)$ in one variable x over k is an intermediate extension of $k((x))/k$.

Exercise 1.15. Let k be a field, and let $k(t)$ be the field of rational functions in one variable t over k. Show that any element of $k(t)$ is a root of some quadratic polynomial with coefficients in $k(t^2)$. Determine all field homomorphisms from $k(t)$ to itself that fix $k(t^2)$ pointwise.

§1.4 Generated subfields

Definition 1.16. Let L/K be a field extension. Let S be a subset of L. The **field generated by S over K**, denoted by $K(S)$, is defined to be the smallest subfield of L that contains both K and S. We say that the field extension L/K is **generated by** S if $L = K(S)$. The **ring generated by S over K**, denoted by $K[S]$, is defined to be the smallest subring of L that contains both K and S.

Remark 1.17. Note that the field generated by S over K is the intersection of all subfields of L that contain both K and S. Similarly, the ring generated by S over K is the intersection of all subrings of L that contain both K and S.

Definition 1.18. Let L/K be a field extension. We say that the extension L/K is **finitely generated** if there exists a finite subset $S \subset L$ such that $L = K(S)$.

Exercise 1.19. Let k be a field. Consider the field extension $k(x_1, x_2, \ldots, x_n)/k$, where $k(x_1, x_2, \ldots, x_n)$ is the field of rational functions in n variables $x_1, x_2, \ldots, x_n$ over k. Show that $k(x_1, x_2, \ldots, x_n)$ is generated by the elements $x_1, x_2, \ldots, x_n$ over

k, that is, show that $k(x_1, x_2, \dots, x_n) = k(\{x_1, x_2, \dots, x_n\})$. Also show that the ring $k[x_1, \dots, x_n]$ of polynomials in n variables $x_1, \dots, x_n$ over k is generated by the elements $x_1, \dots, x_n$ over k, that is, show that $k[x_1, \dots, x_n] = k[\{x_1, \dots, x_n\}]$.

Remark 1.20. The field of rational functions $k(x_1, x_2, \dots, x_n)$ in n variables $x_1, x_2, \dots, x_n$ over k is the field of fractions of the integral domain $k[x_1, \dots, x_n]$ of polynomials in n variables $x_1, \dots, x_n$ over k. Moreover, the field $k(x_1, x_2, \dots, x_n)$ is generated by the elements $x_1, x_2, \dots, x_n$ over k, and hence, the extension $k(x_1, x_2, \dots, x_n)/k$ is finitely generated. Furthermore, the ring $k[x_1, \dots, x_n]$ is generated by the elements $x_1, \dots, x_n$ over k.

Exercise 1.21. Let L/K be a field extension, and let S be a subset of L. Show that the field generated by S over K is equal to the set of all elements of L that can be expressed as a rational function in the elements of S with coefficients in K. In other words, show that

$$K(S) = \bigcup_{\substack{S_f \subset S, \\ |S_f| < \infty}} K(S_f).$$

Also show that the ring generated by S over K is equal to the set of all elements of L that can be expressed as a polynomial in the elements of S with coefficients in K. In other words, show that

$$K[S] = \bigcup_{\substack{S_f \subset S, \\ |S_f| < \infty}} K[S_f].$$

Remark 1.22. Let L/K be a field extension, and let $\alpha_1, \alpha_2, \dots, \alpha_n \in L$. Then the field $K(\{\alpha_1, \alpha_2, \dots, \alpha_n\})$ is also denoted by $K(\alpha_1, \alpha_2, \dots, \alpha_n)$.

Exercise 1.23. Show that the fields $\mathbb{Q}\left(\sqrt{2}\right), \mathbb{Q}\left(\sqrt{3}\right)$ are not isomorphic. What can be said about the fields $\mathbb{Q}\left(\sqrt{2}, \sqrt{6}\right), \mathbb{Q}\left(\sqrt{3}, \sqrt{6}\right)$?

Exercise 1.24. Show that $\mathbb{Q}\left(\sqrt{2}\right)$ is equal to $\mathbb{Q}\left[\left\{\sqrt{2}\right\}\right]$.

Exercise 1.25. Show that $\mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right)$ is equal to $\mathbb{Q}\left(\sqrt{2} + \sqrt{3}\right)$.

Exercise 1.26. Show that $\mathbb{Q}\left(\sqrt{2}, \omega\right)$ is equal to $\mathbb{Q}\left(\sqrt{2} + \omega\right)$, where ω denotes a primitive cube root of unity in $\mathbb{C}$.

Example 1.27. Let n be a positive integer, and let ζ_n denote a primitive n-th root of unity in $\mathbb{C}$. Then the extension $\mathbb{Q}(\zeta_n)$ of $\mathbb{Q}$, generated ζ_n, is called the **cyclotomic extension** of $\mathbb{Q}$ obtained by adjoining the n-th roots of unity.

Exercise 1.28. Determine the degree of the cyclotomic extension $\mathbb{Q}(\zeta_n)$ over $\mathbb{Q}$ for $n = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$.

§1.5 Degree of field extensions

If L/K is a field extension, then L can be viewed as a vector space over K, where scalar multiplication is given by the field multiplication. In particular, we can consider the dimension of L as a vector space over K.

Exercise 1.29. Show that the map

$$\mathbb{C} \times \mathbb{C} \to \mathbb{C}, (a, b) \mapsto \overline{a}b,$$

where $\overline{a}$ denotes the complex conjugate of a, defines a $\mathbb{C}$-linear structure on $\mathbb{C}$ that makes $\mathbb{C}$ into a vector space over $\mathbb{C}$.

Definition 1.30. Let L/K be a field extension. We say that L is a **finite extension** of K if L is a finite-dimensional vector space over K. In this case, we define the **degree** of the extension L/K to be the dimension of L as a vector space over K, and we denote it by $[L : K]$. If L is not finite-dimensional over K, we say that the extension is **infinite**.

Exercise 1.31. Let L/K be a field extension. Show that the degree of the extension L/K is equal to 1 if and only if $L = K$.

Exercise 1.32. Show that $[k(x) : k] = \infty$ for any field k.

Exercise 1.33. Show that $k(t)/k(t^2)$ is a finite extension of degree 2.

Exercise 1.34. Determine the degrees of the following extensions.

$$\mathbb{Q}\left(\sqrt{2}\right)/\mathbb{Q}, \mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right)/\mathbb{Q}, \mathbb{Q}\left(\sqrt{2}, \sqrt[3]{3}\right)/\mathbb{Q}, \mathbb{Q}\left(\sqrt{2}, \sqrt{3}, \sqrt{5}\right)/\mathbb{Q}.$$

Exercise 1.35. Determine the degrees of the following extensions over $\mathbb{Q}$.

$$\mathbb{Q}(i), \mathbb{Q}(\omega), \mathbb{Q}(\zeta_n).$$

Exercise 1.36. Let L/K be a field extension. Show that if L/K is a finite extension of degree n, then any element of L is a root of a polynomial of degree at most n with coefficients in K.

Now let's look at some examples of finite extensions.

Example 1.37. Consider the field extension $\mathbb{Q}\left(\sqrt{2}\right)/\mathbb{Q}$. We claim that this is a finite extension of degree 2. To see this, note that any element of $\mathbb{Q}\left(\sqrt{2}\right)$ can be written in the form $a + b\sqrt{2}$ for some $a, b \in \mathbb{Q}$. The set $B = \left\{1, \sqrt{2}\right\}$ is linearly independent over $\mathbb{Q}$, and it spans $\mathbb{Q}\left(\sqrt{2}\right)$, so it forms a basis. Therefore, the dimension of $\mathbb{Q}\left(\sqrt{2}\right)$ as a vector space over $\mathbb{Q}$ is 2, and we have $\left[\mathbb{Q}\left(\sqrt{2}\right) : \mathbb{Q}\right] = 2$.

Lemma 1.38. Let L/K, K/F be field extensions. If L/K, K/F are finite extensions, then so is L/F, and the degrees of these extensions satisfy

$$[L : F] = [L : K][K : F].$$

Proof. Note that if the degree of the extension K/F is infinite, then observing that K is an F-subspace of L, it follows that the degree of the extension L/F is also infinite. Also note that if the degree of the extension L/K is infinite, then the degree of the extension L/F is also infinite, since any subset of L that generates L as a K-vector space also generates L as an F-vector space. This shows that if either $[L : K]$ or $[K : F]$ is infinite, then $[L : F]$ is also infinite.

Let $\{u_1, u_2, \ldots, u_m\}$ be a basis of L over K, and let $\{v_1, v_2, \ldots, v_n\}$ be a basis of K over F. We claim that the set

$$B = \left\{u_i v_j : 1 \leq i \leq m, 1 \leq j \leq n\right\}$$

is a basis of L over F. Indeed, for any $\alpha \in L$, we can write

$$\alpha = \sum_{i=1}^{m} c_i u_i$$

for some $c_1, \ldots, c_m$ lying in K. Moreover, for each c_i, we can write

$$c_i = \sum_{j=1}^{n} d_{ij} v_j$$

for some d_{ij} lying in F. This shows that

$$\alpha = \sum_{i=1}^{m} \sum_{j=1}^{n} d_{ij} u_i v_j,$$

which shows that B generates L over F. To show that B is linearly independent over F, suppose that we have a linear combination

$$\sum_{i=1}^{m}\sum_{j=1}^{n} e_{ij}u_i v_j = 0$$

for some e_{ij} lying in F. Then we can rewrite this as

$$\sum_{i=1}^{m}\left(\sum_{j=1}^{n} e_{ij}v_j\right)u_i = 0.$$

Since $\{u_1, u_2, ..., u_m\}$ is linearly independent over K, we must have

$$\sum_{j=1}^{n} e_{ij}v_j = 0$$

for each $i = 1, ..., m$. Since $\{v_1, v_2, ..., v_n\}$ is linearly independent over F, we must have $e_{ij} = 0$ for all i, j. This shows that B is linearly independent over F, and hence it is a basis of L over F. Therefore, we have

$$[L : F] = |B| = mn = [L : K][K : F].$$

□

Exercise 1.39. If an extension has prime degree, show that it has no proper intermediate extensions.

Lemma 1.40. Let L/K be a field extension, and let $\alpha_1, \alpha_2, ..., \alpha_n \in L$. Then the extension $K(\alpha_1, \alpha_2, ..., \alpha_n)/K$ is finite if and only if each of the extensions $K(\alpha_i)/K$ is finite for $i = 1, ..., n$. Moreover, if these extensions are finite, then we have

$$[K(\alpha_1, \alpha_2, ..., \alpha_n) : K] \leq \prod_{i=1}^{n}[K(\alpha_i) : K].$$

§1.6 Algebraic elements and algebraic extensions

Exercise 1.41. Let L/K be a field extension, and let $\alpha \in L$.

- Show that the multiplication by α map, given by
$$L \to L, \quad x \mapsto \alpha x,$$
defines a K-linear transformation of L as a vector space over K.
- If L/K is a finite extension, the characteristic polynomial of this linear transformation is a monic polynomial in $K[x]$ of degree $[L : K]$ vanishing at α.
- Assume that α is algebraic over K, and $L = K(\alpha)$. Prove that the characteristic polynomial of this linear transformation is equal to the minimal polynomial of α over K.

Definition 1.42. Let L/K be a field extension. An element $\alpha \in L$ is said to be **algebraic** over K if there exists a non-zero polynomial $f(x) \in K[x]$ such that $f(\alpha) = 0$. If no such polynomial exists, we say that α is **transcendental** over K.

Exercise 1.43. Is it true that algebraic extensions are finitely generated? If yes, prove it. If not, give a counterexample.

Corollary 1.44. Let L/K be a field extension, and let α be an element of L. Then the following statements are equivalent.

- The extension $K(\alpha)/K$ is finite.
- The element α is algebraic over K.

Moreover, α is algebraic over K if and only if α^n is algebraic over K for some positive integer n.

Definition 1.45. Let L/K be a field extension. If an element $\alpha \in L$ is algebraic over K, then the **minimal polynomial** of α over K is the unique monic polynomial $f(x) \in K[x]$ of least degree such that $f(\alpha) = 0$.

Lemma 1.46. Let L/K be a field extension, and let $\alpha \in L$ be algebraic over K. Then the following statements hold.

- The minimal polynomial of α over K is irreducible in $K[x]$.
- If $f(x) \in K[x]$ is any polynomial such that $f(\alpha) = 0$, then the minimal polynomial of α over K divides $f(x)$ in $K[x]$.
- The degree of the extension $K(\alpha)/K$ is equal to the degree of the minimal polynomial of α over K.
- We have

$$K(\alpha) = K[\alpha].$$

Lemma 1.47. Let L/K be a field extension. If L/K is a finite extension, then L is algebraic over K, and it is finitely generated as a field extension of K.

Proposition 1.48. Let L/K be a field extension, and let X be a subset of L. Assume that every element of X is algebraic over K. Then the extension $K(X)/K$ is algebraic. Moreover, if X is finite, then the extension $K(X)/K$ is finite.

Theorem 1.49. Let L/K, K/F be field extensions. If $L/K, K/F$ are algebraic extensions, then so is L/F.

Lemma 1.50. Let L/K be a field extension. The set of all elements of L that are algebraic over K forms a subfield of L, and it is the largest algebraic extension of K contained in L.

Definition 1.51. Let L/K be a field extension. The **algebraic closure** of K in L is the set of all elements of L that are algebraic over K. The field extension L/K is called an **algebraic extension** if every element of L is algebraic over K.

Exercise 1.52. Let $\overline{\mathbb{Q}}$ denote the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$. Show that $\overline{\mathbb{Q}}$ contains an extension of $\mathbb{Q}$ of degree n for every positive integer n. Using this, show that $\overline{\mathbb{Q}}$ is an infinite extension of $\mathbb{Q}$.

§1.7 Compositum of field extensions

Definition 1.53. Let L_1/F and L_2/F be field extensions, contained in a common extension K of F. The **compositum** of L_1 and L_2 over F, denoted by L_1L_2, is defined to be the smallest subfield of K that contains both L_1 and L_2. In other words, L_1L_2 is the intersection of all subfields of K that contain both L_1 and L_2. The compositum L_1L_2 is also called the **composite field** of L_1 and L_2 over F.

Exercise 1.54. If L_1/F and L_2/F are finite extensions contained in a common extension K of F, and if

$$L_1 = F(\alpha_1, \alpha_2, \dots, \alpha_m), \quad L_2 = F(\beta_1, \beta_2, \dots, \beta_n),$$

then show that

$$L_1L_2 = F(\alpha_1, \alpha_2, \dots, \alpha_m, \beta_1, \beta_2, \dots, \beta_n).$$

Exercise 1.55. Show that if L_1/F and L_2/F are finite extensions contained in a common extension K of F, then the compositum L_1L_2/F is also a finite extension of F, and we have

$$[L_1L_2 : F] \leq [L_1 : F][L_2 : F]. \tag{1}$$

Provide an example of two finite extensions L_1/F and L_2/F such that the inequality Equation 1 is strict. Moreover, if $[L_1 : F]$ and $[L_2 : F]$ are coprime, then we have

$$[L_1L_2 : F] = [L_1 : F][L_2 : F].$$

§1.8 Automorphisms of field extensions

Definition 1.56. Let L/K be a field extension. The set of all K-automorphisms of L, denoted by $\mathrm{Gal}(L/K)$, is called the **Galois group** of L over K, and is defined as the set of all field automorphisms of L that fix K pointwise.

Exercise 1.57. Show that $\mathrm{Gal}(L/K)$ is a group under composition of maps.

Lemma 1.58. Let L/K be a field extension, and let X be a subset of L such that $L = K(X)$. Show that any K-automorphism of L is uniquely determined by its action on the elements of X, that is, if σ, τ are K-automorphisms of L such that $\sigma(x) = \tau(x)$ for all $x \in X$, then $\sigma = \tau$.

Lemma 1.59. Let L/K be a field extension, and let α be an element of L that is algebraic over K. Show that any K-automorphism σ of L maps α to a root of the minimal polynomial of α over K, and $\sigma(\alpha)$ is also algebraic over K with the same minimal polynomial as α.

Corollary 1.60. Let L/K be a field extension of finite degree. Then $\mathrm{Gal}(L/K)$ is a finite group.

Exercise 1.61. Determine the Galois group of the following field extensions.

- $\mathbb{Q}\left(\sqrt{2}\right)/\mathbb{Q}$.
- $\mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right)/\mathbb{Q}$.
- $\mathbb{Q}\left(\sqrt{2}, \sqrt[3]{3}\right)/\mathbb{Q}$.
- $\mathbb{Q}\left(\sqrt{2}, \sqrt{3}, \sqrt{5}\right)/\mathbb{Q}$.
- $\mathbb{Q}\left(\sqrt[3]{3}\right)/\mathbb{Q}$.
- $\mathbb{Q}(i)/\mathbb{Q}$.
- $C/\mathbb{R}$.
- $k(t)/k(t^2)$ where k is a field.

Exercise 1.62. Let k be a field.

- Determine whether $k[t]/(t^2 + 1)$ is a field. If it is a field, determine its Galois group over k.
- Determine whether $k[t]/(t^2 + t + 1)$ is a field. If it is a field, determine its Galois group over k.

Remark 1.63. Let K/F be a field extension, and let L be an intermediate extension of K/F. Then the Galois group of K over L is a subgroup of the Galois group of K over F.

Definition 1.64. Let K/F be a field extension, and let L be an intermediate extension of K/F. If H is a subgroup of the Galois group of K over F, then the **fixed field** of H, is denoted by K^H, and is defined to be the set of all elements of K that are fixed by every automorphism in H. For any subset S of $\mathrm{Gal}(K/F)$, we define the **fixed field** of S to be

$$K^S = \{x \in K : \sigma(x) = x \text{ for all } \sigma \in S\}.$$

Exercise 1.65. Show that the fixed field K^H of a subgroup H of the Galois group of K over F is an intermediate extension of K/F. For any subset S of $\mathrm{Gal}(K/F)$, show that the fixed field K^S is equal to the intermediate extension $K^{\langle S\rangle}$ of K/F, where $\langle S\rangle$ denotes the subgroup of $\mathrm{Gal}(K/F)$ generated by S.

Thus far we have seen that the Galois group $\mathrm{Gal}(K/F)$ of a field extension K/F is a group, and the fixed field K^H of a subgroup H of $\mathrm{Gal}(K/F)$ is an intermediate extension of K/F, and we have also seen that the Galois group of K over an intermediate extension L is a subgroup of $\mathrm{Gal}(K/F)$.

Exercise 1.66. Determine the Galois group of the extension $\mathbb{Q}\left(\sqrt{2}, \sqrt[3]{3}\right)/\mathbb{Q}$ and its subgroups, and determine the corresponding intermediate extensions of $\mathbb{Q}\left(\sqrt{2}, \sqrt[3]{3}\right)/\mathbb{Q}$. Are there more intermediate extensions than subgroups of the Galois group?

Exercise 1.67. Are there similar examples in positive characteristic?

Exercise 1.68. Are there more intermediate extensions than subgroups of the Galois group for the extension $\mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right)/\mathbb{Q}$?

Lemma 1.69. Let K/F be a field extension. Then the following statements hold.

- The map
  $\{L : L \text{ is a subextension of } K/F\} \to \{H : H \text{ is a subgroup of } \mathrm{Gal}(K/F)\}$,
  given by
  $$L \rightsquigarrow \mathrm{Gal}(K/L),$$
  is inclusion-reversing, that is, if L_1, L_2 are intermediate extensions of K/F such that $L_1 \subseteq L_2$, then we have

$$\operatorname{Gal}(K/L_2) \subseteq \operatorname{Gal}(K/L_1).$$

- The map

  $$\{L : L \text{ is a subextension of } K/F\} \leftarrow \{H : H \text{ is a subgroup of} \operatorname{Gal}(K/F)\},$$

  given by

  $$K^H \leftarrow\!\!\!\!\rightsquigarrow H,$$

  is also inclusion-reversing, that is, if H_1, H_2 are subgroups of $\operatorname{Gal}(K/F)$ such that $H_1 \subseteq H_2$, then we have

  $$K^{H_1} \supseteq K^{H_2}.$$

- For any intermediate extension L of K/F, we have

  $$L \subseteq K^{\operatorname{Gal}(K/L)}.$$

  In particular, for any subgroup H of $\operatorname{Gal}(K/F)$, we have

  $$K^H = K^{\operatorname{Gal}(K/K^H)}.$$

- For any subset S of $\operatorname{Gal}(K/F)$, we have

  $$S \subseteq \operatorname{Gal}\left(K/K^S\right).$$

  In particular, for any intermediate extension L of K/F, we have

  $$\operatorname{Gal}(K/L) = \operatorname{Gal}\left(K/K^{\operatorname{Gal}(K/L)}\right).$$

Corollary 1.70. Let K/F be a field extension. Then the maps

$$L \rightsquigarrow \operatorname{Gal}(K/L), \quad H \rightsquigarrow K^H$$

provide an inclusion-reversing one-to-one correspondence between the set of all intermediate extensions of K/F which are the fixed fields of subgroups of $\operatorname{Gal}(K/F)$, and the set of all subgroups of $\operatorname{Gal}(K/F)$ which are the Galois groups of K over intermediate extensions of K/F.

Note that the inclusion-reversing one-to-one correspondence in the above lemma is not necessarily a one-to-one correspondence between all intermediate extensions of K/F and all subgroups of $\operatorname{Gal}(K/F)$, since there may be intermediate extensions of K/F that are not fixed fields of any subgroup of $\operatorname{Gal}(K/F)$, and there may be subgroups of $\operatorname{Gal}(K/F)$ that are not Galois groups of K over any intermediate extension of K/F. It is natural to ask when the above inclusion-reversing one-to-one correspondence is a one-to-one correspondence between all intermediate extensions of K/F and all subgroups of $\operatorname{Gal}(K/F)$. A necessary condition for this to happen is that the fixed field of the Galois group $\operatorname{Gal}(K/F)$ is equal to F, that is, we have

$$K^{\operatorname{Gal}(K/F)} = F.$$

This leads to the following definition.

Definition 1.71. Let K be an algebraic extension of a field F. We say that K/F is a **Galois extension** if the fixed field of the Galois group $\mathrm{Gal}(K/F)$ is equal to F, that is, we have

$$K^{\mathrm{Gal}(K/F)} = F.$$

Note that any field automorphism σ of K can be viewed as a group homomorphism from $K^\times$ to itself, where $K^\times$ denotes the multiplicative group of non-zero elements of K.

Definition 1.72. Let G be a group, and let K be a field. A **character** of G over K is a group homomorphism from G to the multiplicative group $K^\times$ of non-zero elements of K. The set of all characters from G to K is denoted by $\mathrm{Hom}(G, K^\times)$.

Lemma 1.73. Let $\sigma_1, \ldots, \sigma_n$ be distinct characters of a group G over a field K. Then the characters $\sigma_1, \ldots, \sigma_n$ are linearly independent over K, that is, if we have a linear combination

$$\sum_{i=1}^{n} c_i \sigma_i = 0$$

for some $c_1, \ldots, c_n \in K$, then we must have

$$c_1 = c_2 = \ldots = c_n = 0.$$

The following result is a consequence of Dedekind's lemma.

Proposition 1.74. Let K/F be a finite extension. Then the Galois group $\mathrm{Gal}(K/F)$ is a finite group, and its order is at most the degree of the extension K/F, that is, we have

$$|\mathrm{Gal}(K/F)| \leq [K : F].$$

Proposition 1.75. Let G be a finite subgroup of the group of automorphisms of a field K. Then the order of G is equal to the degree of the extension K/K^G, that is, we have

$$|G| = \left[K : K^G\right].$$

Consequently, the extension K/K^G is finite and Galois, and has G as its Galois group, that is, we have

$$G = \mathrm{Gal}\left(K/K^G\right).$$

Corollary 1.76. A finite extension K/F is Galois if and only if the order of the Galois group $\mathrm{Gal}(K/F)$ is equal to the degree of the extension K/F, that is, we have

$$|\operatorname{Gal}(K/F)| = [K : F].$$

Corollary 1.77. Let K/F be a field extension, and let $\alpha \in K$ be algebraic over F. Then the order of the Galois group $\operatorname{Gal}(F(\alpha)/F)$ is equal to the number of distinct roots of the minimal polynomial of α over F in $F(\alpha)$. Consequently, the extension $F(\alpha)/F$ is Galois if and only if the number of distinct roots of the minimal polynomial of α over F in $F(\alpha)$ is equal to the degree of the extension $F(\alpha)/F$.

Exercise 1.78. Determine the Galois group of the extension $\mathbb{Q}\left(\sqrt[3]{3}\right)/\mathbb{Q}$ and its subgroups, and determine the corresponding intermediate extensions of $\mathbb{Q}\left(\sqrt[3]{3}\right)/\mathbb{Q}$.

Exercise 1.79. Similar problem as the previous exercise, but for the extension $k(t)/k(t^2)$, or for the extension $k(t)/k(t^p)$ for any prime p and any field k of characteristic p.